

Workload-Aware Microarchitectural Tuning for the Go Benchmark

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Abstract

This study explores the microarchitectural optimization of the SPEC95 **099.go** benchmark under a strict 60-unit area constraint. By leveraging profile-guided optimization to prune the design space, I identified that the workload exhibits zero floating-point utilization and high instruction fetch pressure. The optimization process yielded a **29% performance improvement** over the baseline through asymmetric cache partitioning (1024-set I-Cache). Furthermore, I demonstrate a critical trade-off: while Out-of-Order execution maximizes raw throughput (IPC = 0.93), a minimal In-Order architecture maximizes power efficiency, achieving **2x greater performance-per-watt** than the optimal Out-of-Order configuration.

1. Introduction and Objectives

This analysis focuses on designing an optimal processor architecture for the **go** benchmark under strict area constraints. The optimization problem was formulated to maximize Instructions Per Cycle (IPC) and Area Efficiency subject to a constraint of **60 units**.

1.1 Area Cost Model

The area calculation models distinguish between architecture types:

- **In-Order Area:**

$$Area = W \times 1.8 + ALU_{int} \times 2 + Mult_{int} \times 3 + ALU_{fp} \times 4 + Mult_{fp} \times 5$$

- **Out-of-Order Area:**

$$Area_{OoO} = Area_{InOrder} \times 1.4 + 0.5 \times (RUU + LSQ)$$

Where W is pipeline width, RUU is the Register Update Unit size, and LSQ is the Load/Store Queue size.

1.2 Performance Metrics

The study employs two primary metrics:

1. **IPC (Instructions Per Cycle):** Used for raw performance maximization.
2. **Efficiency (IPC / Watt):** Where Area is used as a proxy for Power, representing instructions per cycle per unit area.

2. Baseline Analysis and Workload Characterization

2.1 Baseline Configuration Analysis

The baseline configuration was established by executing the default SimpleScalar `sim-outorder` simulator.

- Configuration: 4-wide Out-of-Order, 16-entry RUU, 8-entry LSQ.
- Resources: 4 Int ALUs, 1 Int Mult, 4 FP ALUs, 1 FP Mult.
- Base Area: 66.88 units (Exceeds constraint by 11.5%).
- Performance: IPC = 0.9118, CPI = 1.0967.
- Efficiency: 0.0136 IPC per unit area.

Critically, the baseline configuration exceeded the 60-unit area constraint, necessitating optimization to reduce area while maintaining or improving performance.

2.2 Profiling and Characterization

Workload characterization was performed using instruction class profiling to understand the computational demands of go.

Memory Intensity

The workload exhibits memory-intensive behavior with 22,929,995 memory operations representing **28.66% of the 80 million committed instructions**.

- Loads: 16,761,025 (20.95%)
- Stores: 6,168,970 (7.71%)
- Load-to-Store Ratio: 2.7:1

Integer Dominance

Integer computation dominates with 45,236,041 occurrences (56.55%).

Floating Point Usage

The most significant discovery was the **complete absence of floating-point computations** (0.00%) and trap operations (0.00%).

This indicated that the default configuration allocated substantial area (20 units) to entirely unused floating-point resources.

3. Design Space Pruning Strategy

To navigate the optimization space efficiently, I employed a pruning strategy based on the profiling data.

1. Floating Point Elimination

Based on the zero-usage discovery, Floating Point ALUs were reduced from 4 to 1, and FP Multipliers were maintained at the minimum value of 1. This yielded immediate area savings without impacting functional correctness.

2. Integer Multiplier Reduction

Further analysis revealed a heavily skewed integer usage pattern. Integer addition accounts for 32.63% of instructions, while integer multiplication accounts for only 0.07%. This **489:1 ratio** indicated that integer multipliers would experience minimal utilization. Consequently, the integer multiplier parameter space was constrained to {1, 2}.

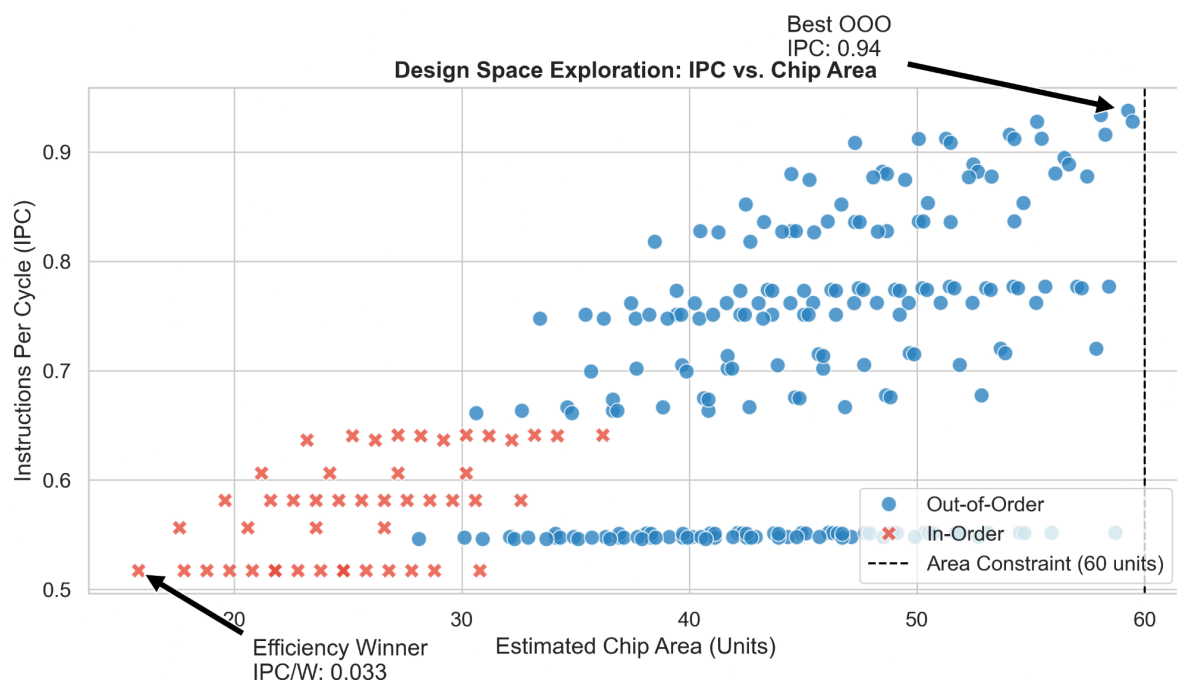
Impact of Pruning

This data-driven approach reduced the Out-of-Order search space from theoretical maximums to just **288 configurations** (Simulation saving: 96.875%). Similarly, for In-Order efficiency runs, the search space was reduced by 93.75%, requiring only 64 simulations.

4. Microarchitectural Optimization Results

4.1 Part A: Maximizing IPC (Out-of-Order)

I explored 209 valid Out-of-Order configurations under the 60-unit limit.



Bottleneck Analysis

A CPI breakdown decomposes the execution cycles.

- IFQ (Instruction Fetch Queue) Full Stalls: 35.22% (indicating fetch bottlenecks).
- RUU Full Stalls: 19.61% (indicating instruction window limitations).
- Branch Misprediction: Contributed to ~18.78% directional inaccuracy.
- LSQ Stalls: Only 1.98%, suggesting minimal memory ordering issues.

Optimal Configuration

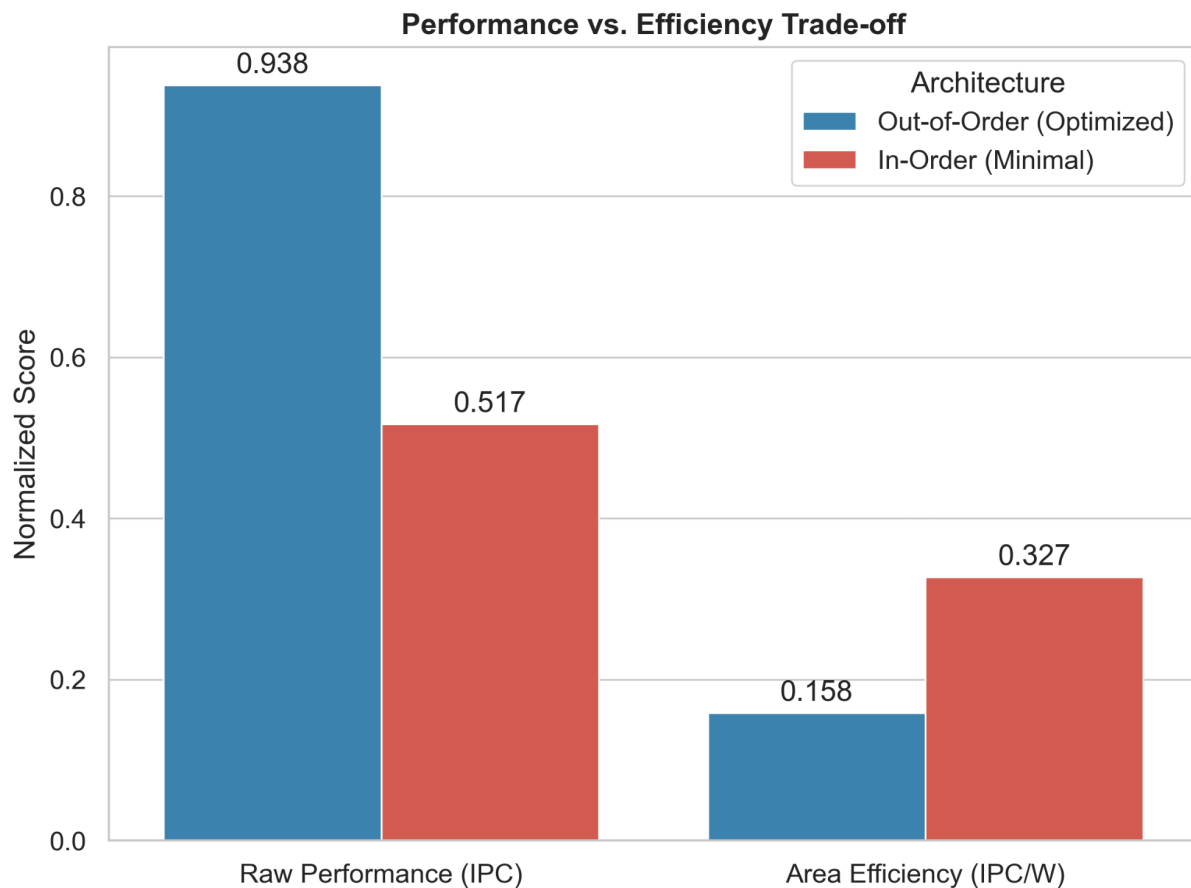
- Params: W=4, IALU=3, IMULT=1, RUU=32, LSQ=16.
- Result: Area = 59.28, **IPC = 0.9376**.

Insight

Performance is gated by instruction-windowing (RUU) and memory reordering. Reducing RUU from 16 to 8 increased penalties severely (from 0.343 to 0.668). The final design prioritizes larger RUU/LSQ sizes over functional unit width.

4.2 Part B: Maximizing Efficiency (In-Order)

To analyze efficiency, I compared the optimized Out-of-Order designs against In-Order architectures.



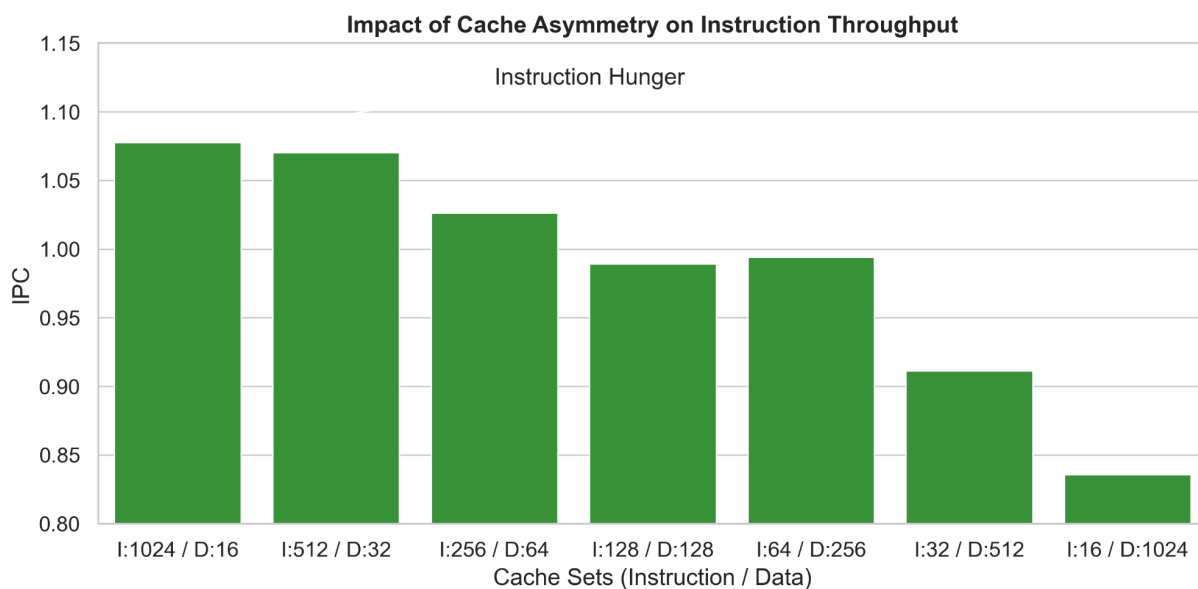
- Optimal Configuration
 - Params: W=1, IALU=1, IMULT=1, In-Order execution.
 - Result: Area = 15.8, IPC = 0.5171, **Efficiency = 0.0327**.
- Trade-off Analysis

In-Order designs consistently surpassed Out-of-Order ones in area efficiency, forming a tight cluster in the higher efficiency band (0.017–0.033).

 - Top In-Order Efficiency: **0.0327**
 - Top Out-of-Order Efficiency: **0.0158**
 - **Conclusion:** The most efficient design doubles the performance-per-watt of the IPC-optimized design, but at the cost of 44.8% lower raw throughput. This confirms that Out-of-Order designs trade efficiency for performance through resource-intensive control structures.

5. Memory Subsystem Optimization (Cache Co-Design)

Given the high Instruction Fetch Queue stall rate (35.22%) identified in Part A, we performed a sensitivity analysis on the L1 Cache hierarchy using the optimal processor configuration.



The analysis revealed a clear hierarchy in performance across seven cache configurations:

- **Best Case (Choice 1):** 1024-set I-Cache / 16-set D-Cache → **IPC = 1.0775**
- **Worst Case (Choice 7):** 16-set I-Cache / 1024-set D-Cache → **IPC = 0.8356**

Analysis:

This outcome underscores an **inverse relationship** between instruction and data cache size requirements for the **go** benchmark. Configurations favoring large instruction caches consistently outperform balanced or data-heavy splits. This indicates that instruction fetch bandwidth and hit rate are far more critical to

pipeline efficiency than data cache size, reflecting a workload with complex control flow but predictable, localized data access patterns.

6. Conclusion

Under a strict 60-unit area budget, this research demonstrates that standard superscalar configurations are inefficient for the `go` benchmark due to unused floating-point resources and instruction fetch bottlenecks.

1. **Profiling-Guided Pruning** and larger RUU/LSQ structures raised the Out-of-Order IPC to **0.9376**.
2. **Cache Co-Design** further raised performance to **1.0775** (a 29% improvement over the worst-case cache config).
3. **Efficiency Analysis** quantified a massive trade-off: a minimal In-Order core offers **2x the efficiency** of the complex Out-of-Order core, providing a clear directional choice depending on whether the target application is high-performance (Server) or power-constrained (Embedded).

Appendix: Reproducibility

To reproduce the results, the following SimpleScalar commands were utilized.

Baseline Simulation:

```
./../simplesim-3.0/sim-outorder -max:inst 80000000 \
-dumpconfig default_config.txt go.ss 50 9 2stone9.in > go.out
```

Workload profiling

```
./../simplesim-3.0/sim-profile -iclass -max:inst 80000000 \
go.ss 50 9 2stone9.in > profile_results.out
```

Stall analysis

```
timeout 300 ./../simplesim-3.0/sim-outorder -max:inst 80000000 \
-fetch:ifqsize 4 -decode:width 4 -issue:width 4 -commit:width 4 \
-res:ialu 4 -res:imult 1 -res:fpalu 4 -res:fpmult 1 \
-ruu:size 16 -lsq:size 8 go.ss 50 9 2stone9.in > stall_stats.out
```